# Data Collection and Preprocessing Phase

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| Date | 25 February 2026 |
| Team ID | 259422 |
| Project Title | Online Payments Fraud Detection using Machine Learning |
| Maximum Marks | 2 Marks |

**Data Quality Report Template**

**<u>High Severity Issues</u>**

Severe class imbalance

Risk of data leakage

These directly impact model evaluation and generalization.

**<u>Moderate Severity Issues</u>**

Skewed distributions

Multicollinearity

Synthetic data characteristics

These influence model interpretability and robustness.

**<u>Low Severity Issues</u>**

Encoding requirement

Limited time resolution.

| Data Source | Data Quality Issue | Severity | Resolution Plan |
|---|---|---|---|
| Dataset | Severe class imbalance (Fraud ≈ 0.13%) | High | Use recall, precision, F1-score instead of accuracy; apply `class_weight='balanced'`; consider SMOTE if required; stratified train-test split. |
| Dataset | Highly skewed transaction amount distribution | Moderate | Apply log transformation for exploratory analysis; retain original values for tree-based models. |

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| Dataset | Strong multicollinearity between balance features (`oldbalanceOrg` & `newbalanceOrig`, `oldbalanceDest` & `newbalanceDest`) | Moderate | Perform correlation matrix analysis; allow tree-based models to handle collinearity; optionally drop one correlated feature for linear models. |
| Dataset | Presence of extreme transaction values (outliers) | Moderate | Validate outliers; retain since they represent potential fraud patterns; avoid trimming unless statistically erroneous. |
| Dataset | Categorical variable (`type`) requires encoding | Low | Apply Label Encoding or One-Hot Encoding depending on model requirements. |
| Dataset | No timestamp granularity (only hourly step) | Low | Treat `step` as numerical proxy for time; no advanced time-series modeling applied. |
| Dataset | Potential synthetic data bias | Moderate | Treat findings as prototype-level insights; avoid overgeneralizing to real-world banking systems. |
| Dataset | Risk of data leakage during preprocessing | High | Ensure target variable excluded from feature engineering; perform scaling after train-test split only. |